Windowpane flounder (*Scophthalmus aquosus*)

Summary Scores

- **Sensitivity:** Low - Moderate
- **Exposure:** Moderate
- **Vulnerability:** Low - Moderate

Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Gulf of Maine/Georges Bank Score	Gulf of Maine/Georges Bank LH DQ	Gulf of Maine/Georges Bank Tally	Southern New England/Mid-Atlantic Score	Southern New England/Mid-Atlantic LH DQ	Southern New England/Mid-Atlantic Tally
Adult Mobility	1.6	2.46		1.6	2.46		1.6	2.46	
Complexity in Reproductive Strategy	2.5	1.98		2.6	1.98		2.4	1.98	
Dispersal of Early Life Stages	2.28	1.1		2.28	1.1		2.28	1.1	
Egg and Larval Survival and Settlement Requirements	1.76	1.71		1.76	1.71		1.76	1.71	
Historical Variability in Temperature	1.16	2.86		1.16	2.86		1.16	2.86	
Juvenile Habitat Requirements	1.84	2.66		1.84	2.66		1.84	2.66	
Other Stressors	1.96	2.31		1.96	2.31		1.96	2.31	
Population Growth Rate	1.48	1.78		1.48	1.78		1.48	1.78	
Prey Specificity	2.08	2.5		2.08	2.5		2.08	2.5	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.12	1.38		1.12	1.38		1.12	1.38	
Spawning Cycle	1.78	2.8		1.88	2.8		1.68	2.8	
Stock Size/Status	2.98	2.85		3.52	2.85		2.44	2.85	
Total Sensitivity | Certainty	**Moderate (2) | 0.52**			**Moderate (2) | 0.69**			**Low (1) | 0.74**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Gulf of Maine/Georges Bank Mean	Gulf of Maine/Georges Bank SD	Gulf of Maine/Georges Bank Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Bottom Temperature	2.62	0.99		2.66	0.99		2.63	0.98	
Bottom Oxygen	2.83	0.99		3.04	0.93		2.69	1.00	
Bottom Salinity	2.89	0.84		2.74	1.06		3.06	0.53	
Bottom Aragonite Solubility	1.08	0.40		1.14	0.54		1.03	0.21	
Small zooplankton integrated biomass	1.35	0.37		1.29	0.29		1.39	0.41	
Medium zooplankton integrated biomass	1.32	0.35		1.25	0.28		1.38	0.39	
Large zooplankton integrated biomass	1.18	0.33		1.15	0.29		1.17	0.31	
Total Exposure	**Moderate (2.13)**	**0.92**		**Moderate (2.24)**	**0.93**		**Moderate (2.06)**	**0.91**	

Vulnerability

Vulnerability	Range Mean	Range SD	Range Tally	Gulf of Maine/Georges Bank Mean	Gulf of Maine/Georges Bank SD	Gulf of Maine/Georges Bank Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Total Vulnerability	**Low (3.15)**	**1.84**		**Moderate (4.44)**	**1.85**		**Low (2.06)**	**0.91**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Windowpane flounder. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 6 and Figure 5). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low - Moderate	None - Increase
Exposure	High	Moderate	Decrease
Vulnerability	Low	Low - Moderate	None - Increase
Directional Effect	Neutral	Negative - Neutral	Decrease - None

Stocks

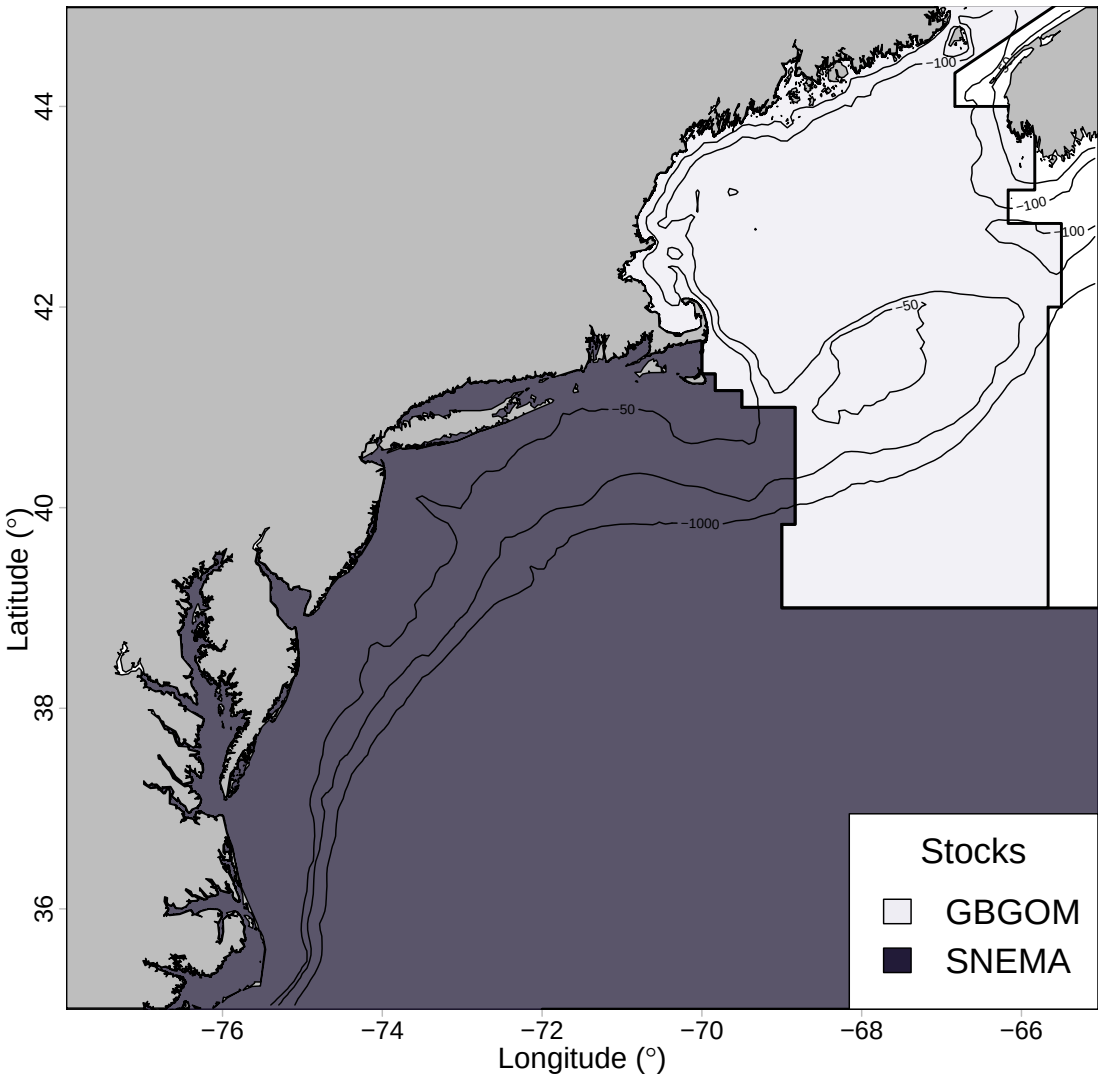


Figure 2: Windowpane flounder Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

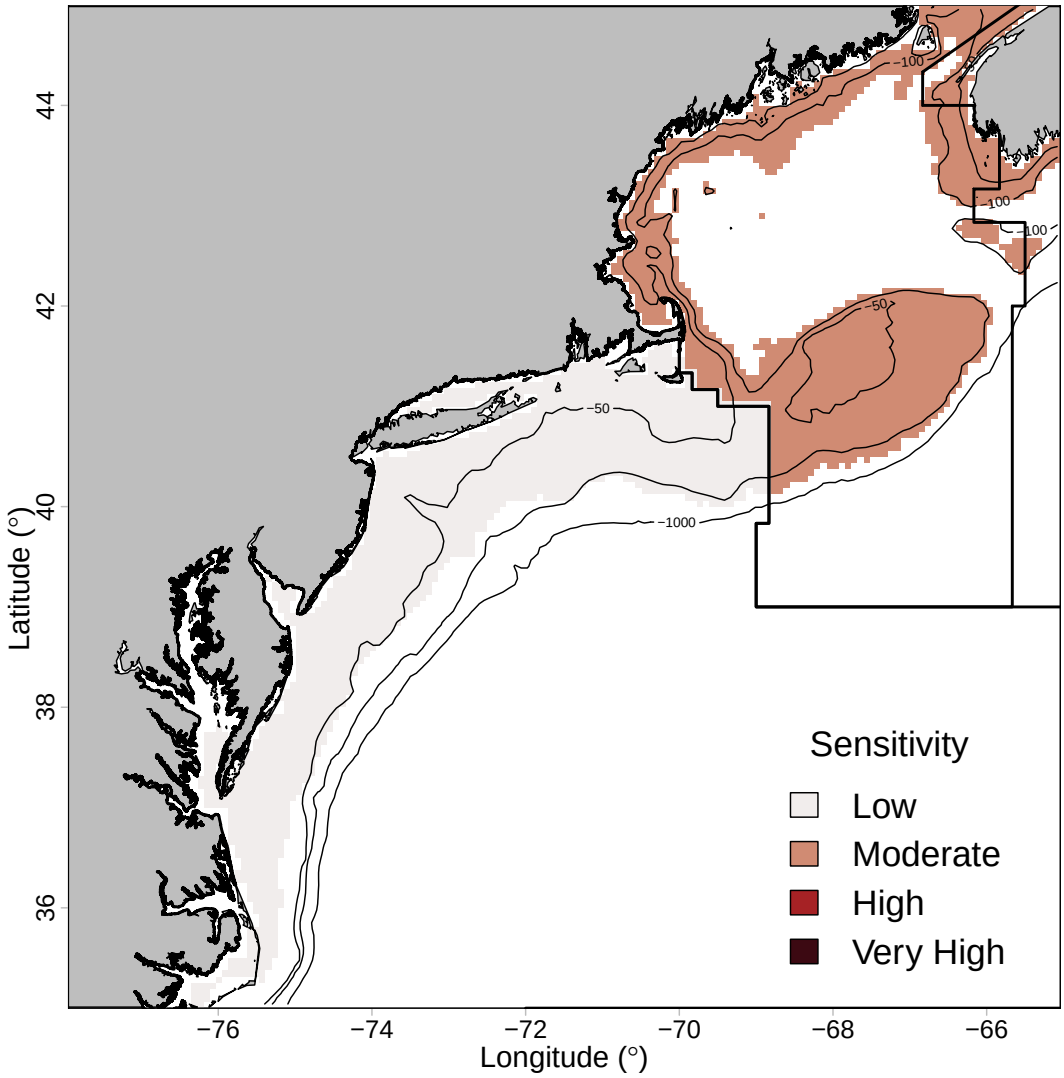


Figure 3: Windowpane flounder Sensitivity on the Northeast Shelf

The **Gulf of Maine/Georges Bank** stock of Windowpane flounder received a **Moderate** sensitivity score, with a certainty of 69%. This score was driven by Stock Size (3.5) and Complexity in Reproductive Strategy (2.6) (Figure 1).

The **Southern New England/Mid Atlantic** stock received a **Low** sensitivity score, with a certainty of 74%. No attributes received a score greater than 2.5 for this stock. Stock Size and Complexity in Reproductive Strategy both received scores of 2.4 (Figure 1).

Spawning occurs throughout most of the year (Able and Fahay 2010) (Cadrin et al. 2021), but relies on cold bottom temperature (8.5-13.5 C) (Chang et al. 1999). Spawning occurs in estuaries and nearshore environments; estuarine water column habitats are considered highly vulnerable (Guyant et al. 2022). The stock status of the Gulf of Maine/Georges Bank stock is unknown. The southern stock is not overfished

and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2023a) (Northeast Fisheries Science Center (NEFSC) 2023b)).

In the previous Climate Vulnerability Assessment, Windowpane flounder received a Low sensitivity score. This was driven by Stock Status (Hare et al. 2016).

Exposure:

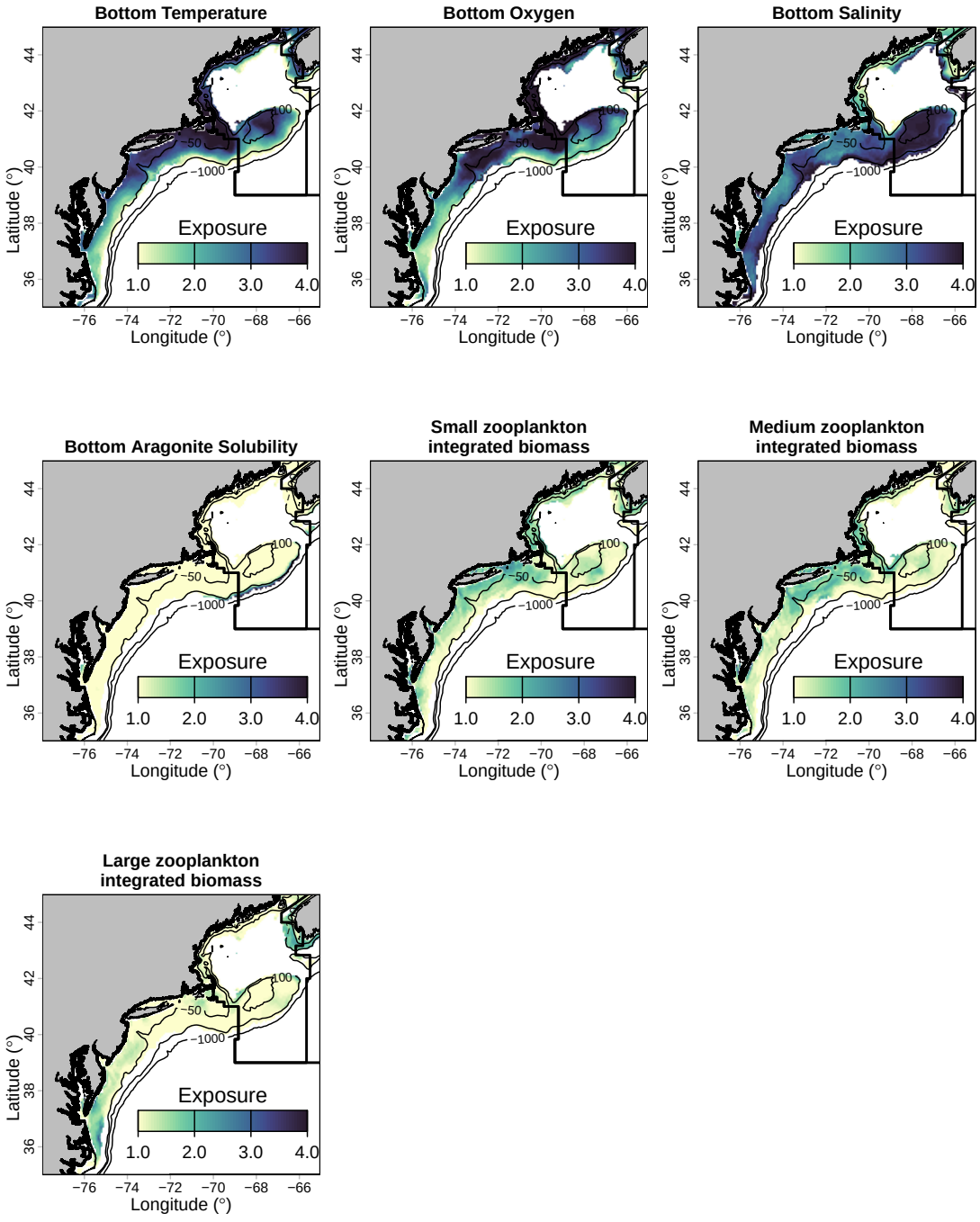


Figure 4: Maps of each Exposure Factor for Windowpane flounder. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Windowpane Flounder was **Moderate** across its range (2.13 +/- 0.92). Scores were also **Moderate** within the **Gulf of Maine/Georges Bank** and **Southern New England/Mid-Atlantic**

stocks (2.24 +/- 0.93 and 2.06 +/- 0.91, respectively) (Figure 1). Exposure was Very High over Georges Bank and High along the coasts of the Gulf of Maine and in Southern New England (Figure 5). These patterns were driven by bottom temperature and oxygen exposure (Figure 4).

In the previous Assessment, Windowpane Flounder received a High exposure score, driven by high sea surface temperature and ocean acidification exposures (Hare et al. 2016).

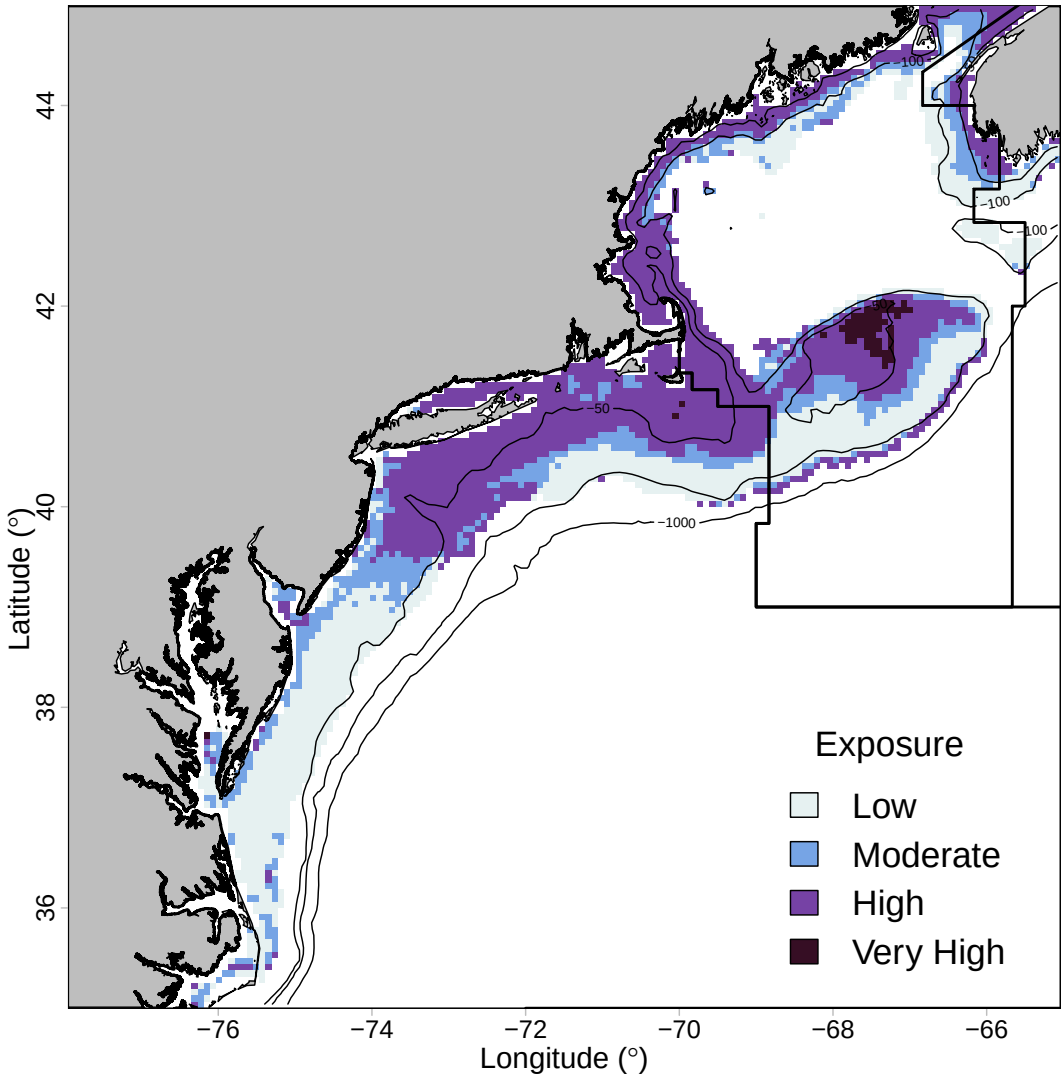


Figure 5: Total Exposure on the Northeast Shelf for Windowpane flounder. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 4)

Vulnerability:

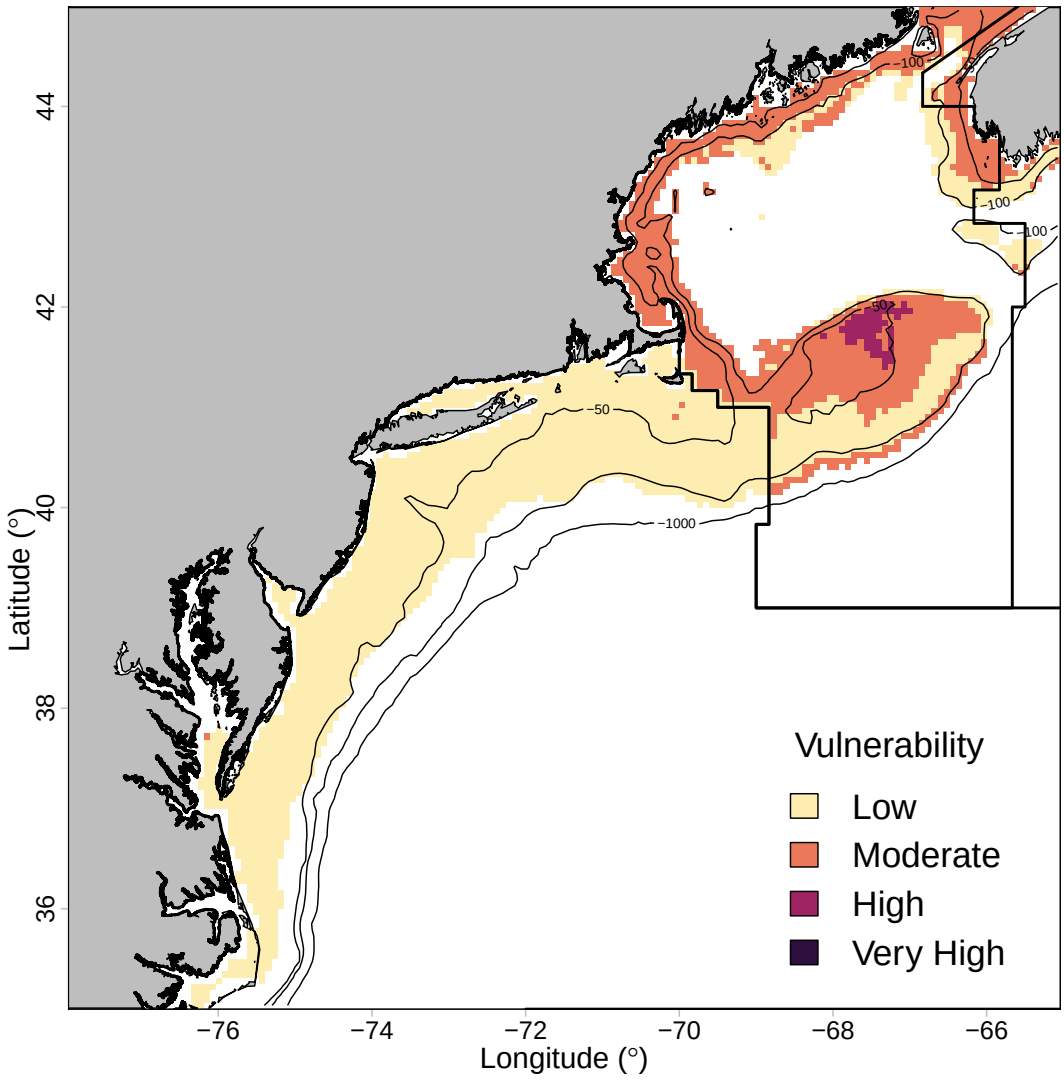


Figure 6: Vulnerability on the Northeast Shelf for Windowpane flounder. Vulnerability is the product of sensitivity and exposure (see Figure 5 and Figure 3)

Mean vulnerability for Windowpane Flounder was **Low** on average across its range (3.15 +/- 1.84) (Figure 1). Average scores were **Moderate** within the **Gulf of Maine/Georges Bank** stock and **Low** within the **Southern New England/Mid Atlantic** stock (4.44 +/- 1.85 and 2.06 +/- 0.91, respectively) (Figure 1). Scores were High over Georges Bank and Moderate along the coast of the Gulf of Maine. Scores were Low throughout the Southern New England/Mid Atlantic stock due to the Low Sensitivity score (Figure 6).

In the previous Assessment, windowpane flounder received a Low vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

75% of data quality scores were greater than or equal to 2, indicating good data quality. Across both stocks, Complexity in Reproductive Strategy, Dispersal of Early Life Stages, Egg and Larval Survival and Settlement Requirements, Population Growth Rate, and Relationship to Calcified Organisms were all identified as data gaps (mean attribute data quality score < 2) (Figure 1).

Model Confidence Score:

The ensemble windowpane flounder distribution model received a 2.1 (+/- 0.2) model confidence score, indicating moderate confidence in the model. The predicted distributions matched experts' expectations. Residuals were higher in nearshore areas, which drove down scores, but were overall evenly distributed. The ensemble model AUC score was 0.92, indicating excellent model performance.

Additional Metrics

Distribution Shifts:

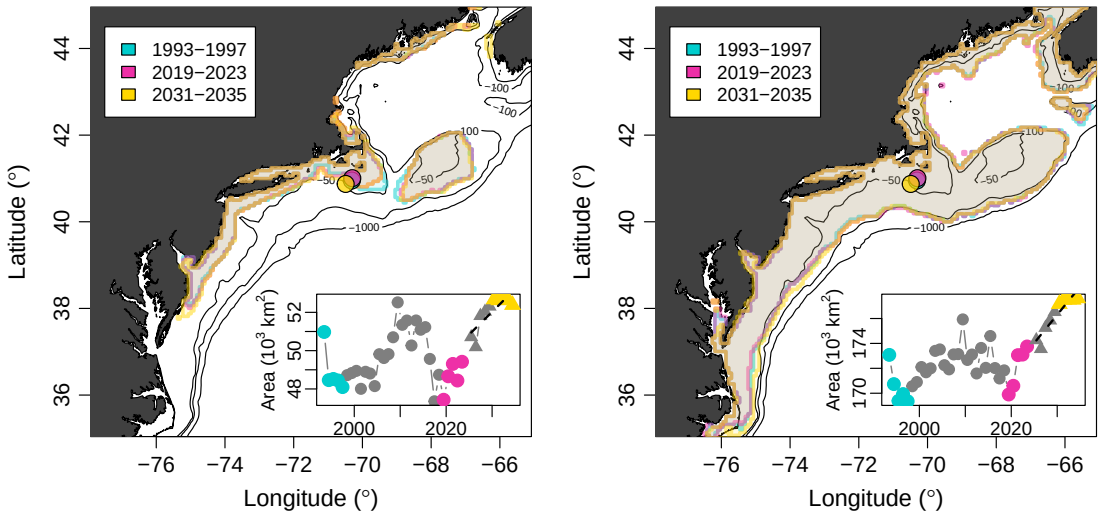


Figure 7: Distribution shifts on the Northeast Shelf for Windowpane flounder. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Windowpane flounder habitat area and range did not change significantly between 1993 and 2023, due to recent declines in habitat area after a period of growth between approximately 2005 and 2015 (Figure 7). During this time, the weighted centroid shifted 6 km N.

Under near future ocean conditions, core habitat and the range of the windowpane flounder are predicted to expand. Core habitat is predicted to extend south along the coast of Maryland and Virginia and their range is expected to expand offshore towards the continental shelf break (Figure 7). The weighted centroid is predicted to shift 22 km southwest.

In the previous Assessment, windowpane flounder received a High Distributional Vulnerability rank due to their seasonal migrations, varied habitat preferences, and high larval dispersal (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	48,997	170,832	6.48	8.90
2019-2023	6.0 km N	48,735 (-262)	172,242 (+1,410)	6.48 (0)	8.90 (0)
2031-2035	21.9 km SW	52,627 (+3,891)	177,465 (+5,223)	6.80 (+0.31)	9.06 (+0.16)

Directional Effect:

Experts predicted that effects of climate change would be **neutral** for the Gulf of Maine/Georges Bank stock, with 73% certainty. For the Southern New England/Mid-Atlantic stock, the effects were predicted to be **negative**, with 58% certainty. Increasing temperatures have been shown to displace windowpane flounder, and extreme cold winters have increased juvenile mortality (Bell et al. 2021) (Wilber et al. 2016). In the previous Climate Vulnerability Assessment, experts predicted that impacts of climate change on windowpane flounder would be neutral (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Percent biomass of windowpane flounder in the Northwest Atlantic increases with changes in seasonality, from cold springs, to warm springs, to fall (Dunn et al. 2016). Increasing temperatures ($>$ 7-10°C) have been shown to displace windowpane flounder (Bell et al. 2021). In New York/New Jersey Harbor, cold winter temperatures decreased windowpane abundance and overwinter mortality in small juveniles (Wilber et al. 2016). Overwintering success appears to be driven in part by temperatures, with extreme cold conditions resulting in high overwinter mortality of age-0 individuals. This likely has a larger impact on fall-spawned juveniles since resilience to extreme cold temperatures is size-dependent and fall-spawned individuals have less time to grow (Neuman and Able 2009). Individual windowpane have also demonstrated avoiding hypoxic conditions (Cadrin et al. 2021).

Life History Synopsis

Windowpane flounder inhabit shallow waters from the Gulf of St. Lawrence to Florida, but are most abundant from Georges Bank to Chesapeake Bay (Chang et al. 1999). They reach maturity around 1.5 years of age and can live up to 8 years (Cadrin et al. 2021). Windowpane spawn from February to November on the inner continental shelf, Georges Bank, and within estuaries. Spawning occurs in the Mid-Atlantic Bight in both spring and fall, whereas spawning in Georges Bank peaks in summer (Able

and Fahay 2010) (Cadrin et al. 2021). Windowpane spawn in the evening or at night on or near bottom at temperatures ranging from 6-21°C (Chang et al. 1999). Eggs hatch after 8 days. The duration of the larval stage is unknown, but eggs and larvae remain in the water column feeding on suspended copepods (Chang et al. 1999). Surveys from Canadian waters suggest that small (< 10 cm TL) windowpane consume bay shrimp and crab larvae, which may be a transition point from zooplankton prey to larger prey items (Hanson and Wilson 2014). Larvae settle to the bottom at approximately 10 mm TL. However, individuals collected on Georges Bank may be planktonic up to 20 mm. There is no indication that settlement is driven by environmental factors (Morse and Able 1995). Juveniles that settle in shallow inshore waters move to deeper offshore waters as they grow. Juveniles inhabiting Georges Bank (< 60 m) undergo seasonal movements to deeper waters along the southern flank of the Bank occur during late autumn, as bottom temperatures drop, and overwintering occurs in deeper areas until late spring (Klein-MacPhee 2002). From Massachusetts to the Gulf of Maine, they can be found in estuaries and on the shelf (Chang et al. 1999). Adults are found on the benthos in waters < 100 m and tolerate a wide range of temperatures (0-26.8°C) (Chang et al. 1999). Adult windowpane may travel along the coast for considerable distances. Juveniles and adults may migrate to nearshore or estuarine habitats in the Mid-Atlantic Bight. Seasonal movements into offshore overwintering habitats have also been documented (Chang et al. 1999). Stomach content data collected during Northeast Fisheries Science Center (NEFSC) bottom trawl surveys indicate windowpane feed on small crustaceans (e.g., mysids and decapod shrimp) and various fish larvae including hakes and tomcod, as well as their own species (Langton and Bowman 1981) (Chang et al. 1999). Windowpane flounder are managed by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. The fishery is divided into two stocks: Northern Gulf of Maine/Georges Bank and Southern New England/Mid-Atlantic. Overfishing and overfished status of the northern stock is unknown, as the assessment has not estimated reference points since 2020 (Northeast Fisheries Science Center (NEFSC) 2023a). The southern stock is not overfished and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2023b).

Supplemental Scores

Exposure Timeseries

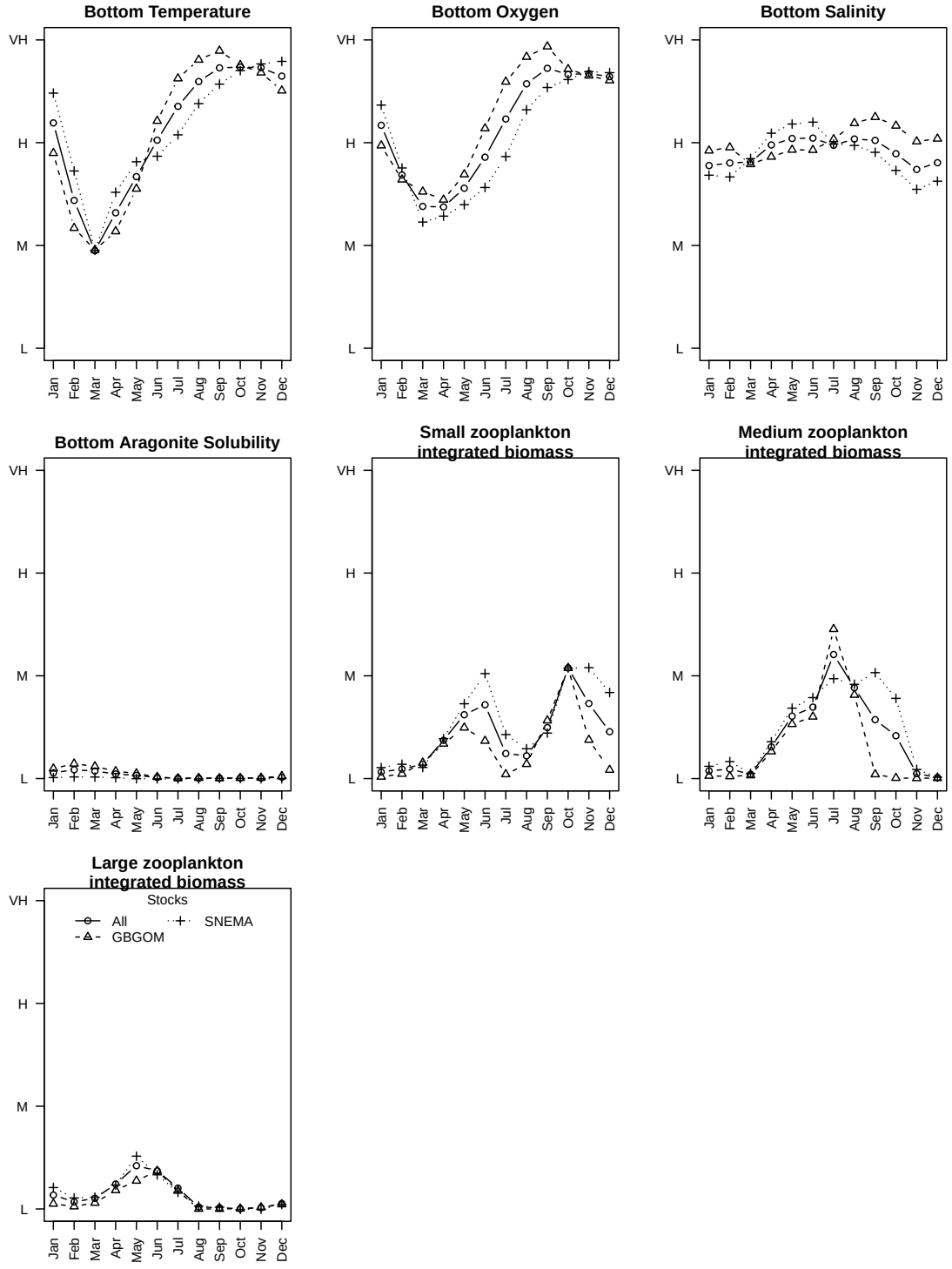


Figure 8: Monthly Timeseries of each Exposure Factor for Windowpane flounder. Exposure was calculated between 2014-2023 and 2025-2035

Exposure varied from Low to High across months. Exposures were Low in early spring (March-April) due to declines in bottom temperature and oxygen exposure (Figure 9, Figure 8).

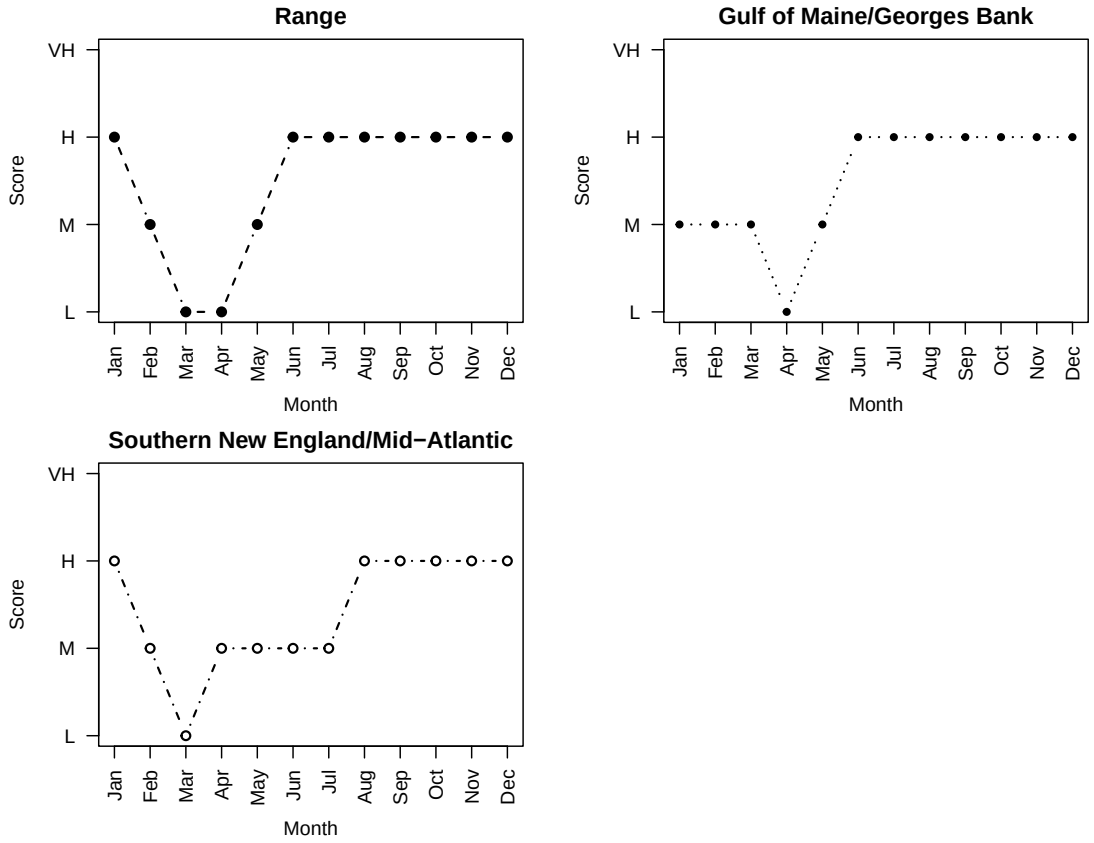


Figure 9: Timeseries of Total Exposure for Windowpane flounder

Exposure due to Significant Factors

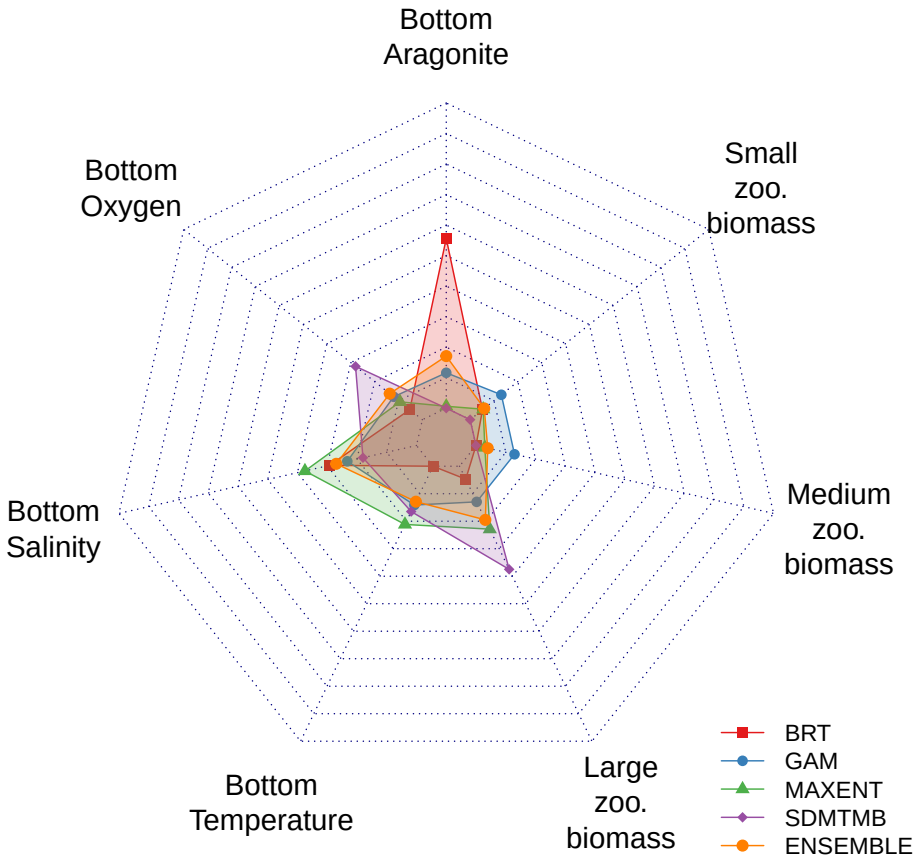


Figure 10: Dynamic Factors driving Windowpane flounder distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, salinity, and aragonite solubility, as well as large zooplankton biomass were important drivers of windowpane distributions (Figure 10). Temperature is a known driver of movement (Klein-MacPhee 2002). Coastal and estuarine habitats may be important overwintering habitats (Chang et al. 1999). Small crustaceans and small fish larvae are the predominant prey species (Langton and Bowman 1981) (Chang et al. 1999). When only these factors were used to calculate exposure, scores remained Moderate both across the range and within stocks (Figure 12; Figure 11).

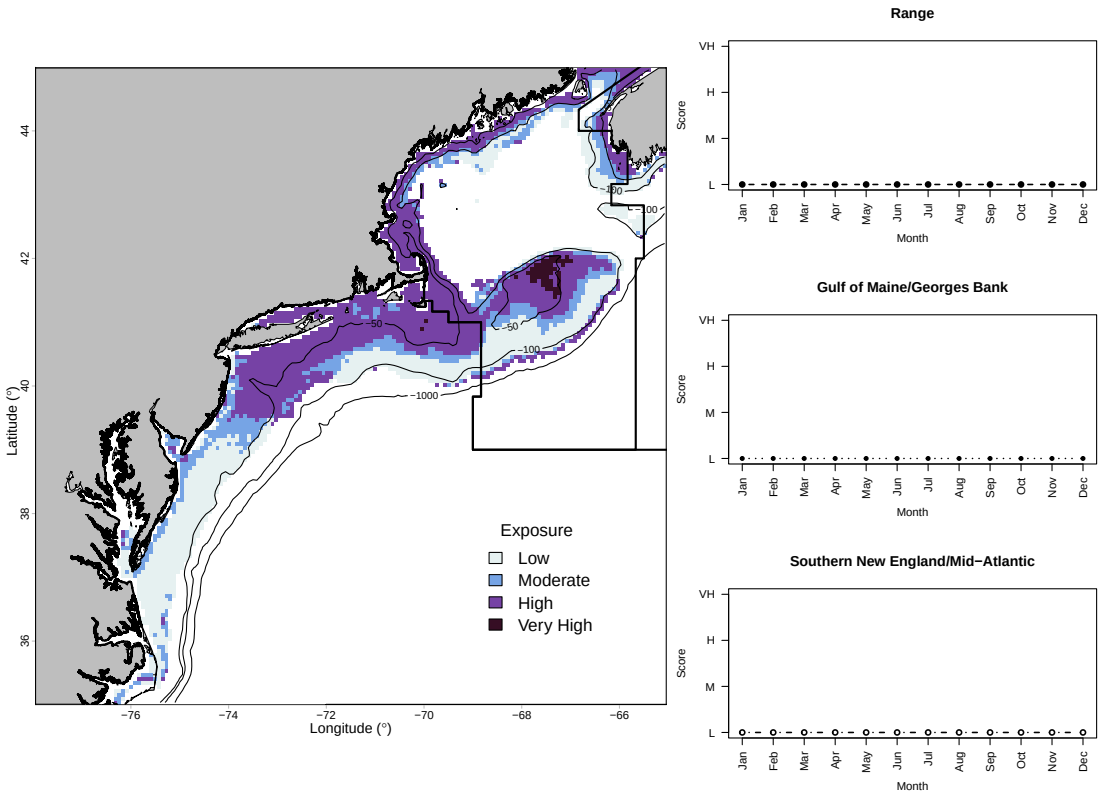


Figure 11: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Windowpane flounder distributions

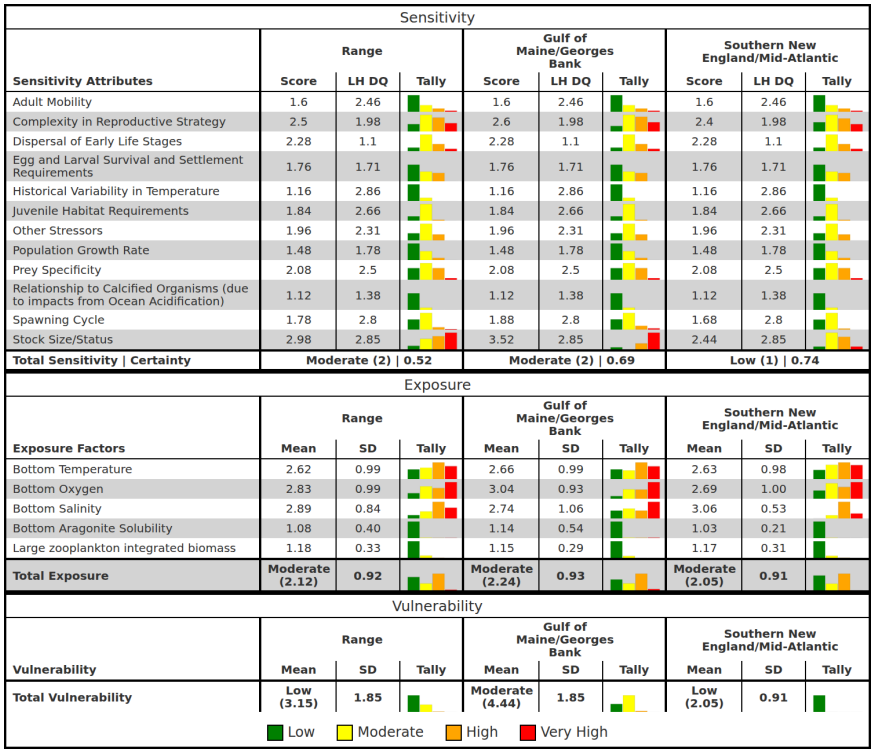


Figure 12: As in Figure 1 for Windowpane flounder, but with only significant Exposure factors and the resulting Vulnerability Scores.

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